

## Propuesta para realizar Proyecto de Grado durante el semestre 2026-2

### 1 Project title:

|                                                                                                                                        |
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| Ridge Formation at Ice-Exposing Scarps on Mars and Earth: Insights from Geomorphological Mapping and Modeling of Sublimation Over Time |
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### 2 Submission date of the project proposal:

|             |
|-------------|
| May 7, 2026 |
|-------------|

### 3 Student information:

|                                      |                   |
|--------------------------------------|-------------------|
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### 4 Information of the director / co-director (in case there is one):

|                                              |                                         |
|----------------------------------------------|-----------------------------------------|
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### 5 Introduction:

The mid-latitude regions of Mars have abundant evidence for the presence of water in the form of shallowly buried ice deposits. The mass balance of these ice sheets is highly relevant to the planet's geomorphological evolution, as they influence the distribution of debris and sedimentary processes ([Baker et al., 2024](#)). These ice sheets undergo both accumulation and sublimation, two factors which vary over time in response to changes in Mars' orbital parameters, namely obliquity, eccentricity, and longitude of perihelion. Analogous to Earth's Milankovitch cycles, these periodic variations are a product of gravitational interactions with

other planetary bodies and induce changes in both the angle and intensity of solar radiation received across the surface of the planet. As a result, the latitudinal zones where ice remains stable vary throughout time.

Unlike Earth, where obliquity fluctuates modestly between  $21^\circ$  and  $24.5^\circ$ , Mars exhibits drastic tilt variations ranging from  $15^\circ$  to  $45^\circ$  over cycles of approximately 120,000 years (Laskar et al., 2004). Even Earth’s comparatively mild changes in tilt are enough to cause ice ages. Mars is therefore expected to undergo extreme ice ages, driven in large part by the amplified obliquity variations. Consequently, the zones of long-term ice stability migrate equatorward under high obliquity conditions, and poleward when the obliquity is low due to variations in the solar incidence angle (Head et al., 2003). On the other hand, the planet’s eccentricity and location of perihelion also vary over time, with cycles of about 96,000 years and 51,000 years, respectively; these periodicities are shown in Figure 1. Variations in eccentricity and location of perihelion lead to differences in solar radiation between perihelion and aphelion, intensifying seasonal temperature extremes. When perihelion aligns with one hemisphere’s summer, that hemisphere gets hotter, shorter summers, while the opposite hemisphere has cooler, longer summers that favor ice accumulation.

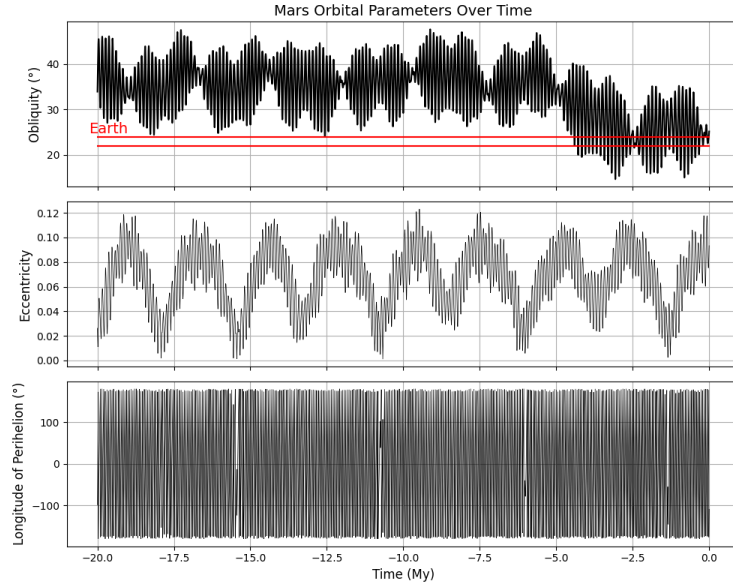


Figure 1: Martian obliquity (compared to Earth’s), eccentricity and longitude of perihelion over the past 0.5Mya.

The influence of these three orbital parameters (obliquity, eccentricity, and longitude of perihelion) is known as orbital forcing, and their combined effect control the total cumulative solar insolation. Changes in insolation driven by orbital forcing impact the stability of mid-latitude ice. As ice sheets sublimate and retreat, they form scarps (cliffs that mark the boundary of the remaining ice). Ice-exposing scarps were discovered in the northern and southern midlatitudes, revealing the internal structure of thick ( $\approx 100$  m), massive ice deposits which may be part of the latitude-dependent mantle (Dundas et al., 2018). These scarp sites are characterized by a steep ( $\approx 45^\circ$ ) pole-facing scarp that is often adjacent to a pit with roughly triangular appearance (Williams et al., 2022) as illustrated in the left image of Figure 2. They consist of a surface layer of lag covering the ice, with additional regolith accumulating at their base. As the scarp retreats, ridge structures would form parallel to it, likely representing cycles in which as sublimation rates speed up and slow down due to variations in orbital forcing.

These scarps are hypothesized to form due to sublimation of the ice sheet initiating at a localized point, after which the scarp retreats equatorward, resulting in a face that is nearly perpendicular to the direction of retreat (Williams et al., 2022). Ridge structures within the pit are observed parallel to the present-day scarp (as shown in Figure 2), which we hypothesize represent previous positions of the scarp.

Similar features have also been identified in terrestrial environments. In places such as the McMurdo Dry Valleys of Antarctica and Alaska, icy cliffs have been observed retreating over time, providing a use-

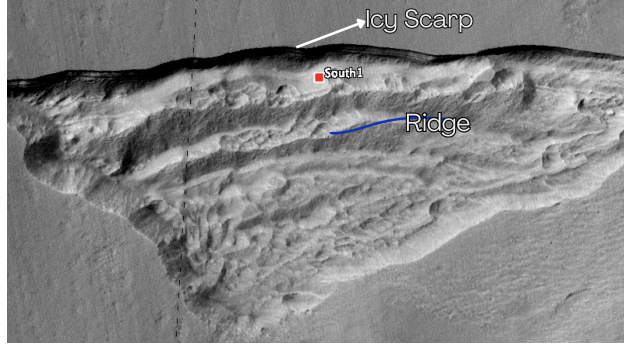


Figure 2: The image shows a HiRISE image of South 1, where both the icy scarp and an adjacent ridge are visible.

ful comparative analogue to Martian icy scarps. Although the environmental conditions and sublimation regimes differ between Mars and Earth, both settings may preserve landforms shaped by ice loss and scarp retreat. These comparisons may therefore provide insight into how sublimation-related processes influence the geomorphology of icy landscapes under different climatic conditions.

Can orbital-forcing-driven variations in sublimation explain ridge formation and retreat at Martian mid-latitude icy scarps, and can these processes be used as a simplified analogue for icy scarps on Earth? In this study we aim to evaluate whether orbital forcing-driven climate variations can explain the spatial patterns of scarp retreat and ridge development observed at Martian mid-latitude icy scarps. To address this, we will combine geomorphologic characterization of ridge morphology and distribution with sublimation modeling for multiple scarp sites in the southern hemisphere to determine whether cyclic variations in ice loss are sufficient to generate the ridge sequences identified in satellite data. In addition, we will analyze possible terrestrial analogues (Alaska and Antarctica) for the martian icy scarps, to determine if the processes which form and evolve these features are analogous to those occurring on Mars.

## 6 General objective

Investigate the relative influence of orbital forcing, sublimation-driven scarp retreat, and ridge-forming processes on the long-term evolution of mid-latitude icy scarps in the southern hemisphere of Mars, by comparing their morphology across multiple sites and with terrestrial analogues, in order to understand how ice-sheet margins change over time and identify indicators of past ice extent and age.

## 7 Specific objectives

- Map and measure selected mid-latitude icy scarps and their ridges in the southern hemisphere of Mars using HiRISE imagery and topographic data.
- Calculate changes in sublimation rates throughout the past 0.5My due to variations in orbital forcing using a one-dimensional thermal model of the martian surface.
- Correlate the cycles of ice sublimation rate variations with the ridges seen in topographic profiles of each scarp according to the horizontal retreat and volume of sublimated ice.
- Use Earth analogues to identify morphologic indicators for previous positions and timing of the ice deposits.
- Analyze terrestrial icy scarp analogues by characterizing its retreat patterns through comparison with Martian icy scarps.

## 8 Methodology

1. Literature review.
2. Martian scarp mapping using JMARS software and HiRISE images.
3. Scarp site characterization using topographic profiles (ridge count and distance measurements).
4. Sublimation rate modeling with a 1D thermal model for Martian surface conditions.
5. Terrestrial analog study using cartography and time series analysis obtained from ground-based and airborne LiDAR.
6. Compare retreat patterns between Martian and Terrestrial scarp sites.

## 9 Budget

When preparing this section, you must take into account the criteria established in the current Geosciences Undergraduate Thesis Regulations. You must have previously approved funding sources, which should be discussed in advance with your advisor(s), who will guide you in this matter. Be as specific as possible. If the proposed project does not require a budget, please write “NOT APPLICABLE”.

NOT APPLICABLE

## 10 Project Schedule

(Adjusted to the 16 weeks of the academic semester. Remember not to schedule work during recess week.)

This schedule should include as many activities as you plan to carry out during the development of your project. List the activities and mark with an X the week(s) in which each activity will be performed.

Friday, September 25 (Week 8): deadline to submit the 30% progress report.

SR: Recess week.

Friday, October 9 (Week 9): deadline for the 30% grade to be uploaded to the system.

| Activities/Week                                                                                                                                                                         | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | SR | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|----|---|----|----|----|----|----|----|----|
| Activity #1: Literature review of previous studies on the retreat and ridge formation in martian icy scarps and the retreat processes of scarp site analogues in Antarctica and Alaska. | X | X | X | X | X | X | X | X |    | X | X  |    |    |    |    |    |    |
| Activity #2: Scarp site mapping across hemispheres using the list of documented scarp sites by <a href="#">Dundas et al. (2021)</a> .                                                   |   | X | X |   |   |   |   |   |    |   |    |    |    |    |    |    |    |
| Activity #3: Analyze topography and elevation profiles at icy scarp sites identified for detailed follow-up.                                                                            |   |   | X | X |   |   |   |   |    |   |    |    |    |    |    |    |    |
| Activity #4: Thermal modeling of martian surface conditions to calculate the sublimation rates until cumulative amount matches total retreat measurements (expected to be 0.5 Myr).     |   |   |   |   | X | X | X |   |    |   |    |    |    |    |    |    |    |
| Activity #5: Qualitative correlation of cycles of ice retreat and ridge formation between the horizontal retreat obtained by the model and the location of the ridges                   |   |   |   |   |   |   | X | X |    |   |    |    |    |    |    |    |    |
| Activity #6: Volume of ice loss calculations for each cycle previously identified, based on the morphology of each scarp site.                                                          |   |   |   |   |   |   |   | X |    | X |    |    |    |    |    |    |    |

|                                                                                                                                                                        |   |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|--|---|---|---|---|---|---|---|---|
| Activity #7: Investigate possible relationships between sublimated ice volume, horizontal retreat and sublimation rate changes to evaluate the initial hypothesis.     |   |   |   |   |   |   |   |   |  | X | X |   |   |   |   |   |   |
| Activity #8: Map scarps and ridges and their changes over time series data for the terrestrial sites (Antarctica and Alaska).                                          |   |   |   |   |   |   |   |   |  |   |   | X |   |   |   |   |   |
| Activity #9: Compare scarp sites and also analyze how they change over time to investigate how ridge formation and retreat may be related at analog sites on Earth.    |   |   |   |   |   |   |   |   |  |   |   | X | X | X |   |   |   |
| Activity #10: Preparation of the written report (introduction, literature review, methodology and preliminary results) and presentation of the 30% progress milestone. | X | X | X | X | X | X | X | X |  |   |   |   |   |   |   |   |   |
| Activity #11: Integration of the information obtained from martian and terrestrial icy scarp analysis.                                                                 |   |   |   |   |   |   |   |   |  |   |   |   | X | X | X | X |   |
| Activity #12: Meetings with director and co-director                                                                                                                   | X |   | X |   | X |   | X |   |  | X |   | X |   | X |   | X |   |
| Activity #13: Preparation of the final written report.                                                                                                                 | X | X | X | X | X | X | X | X |  | X | X | X | X | X | X | X | X |

## 11 Bibliography

- Baker, D., Whelley, P., Richardson, J., Shoemaker, E., Johnson, A., and Matiella Novak, A. (2024). Buried ice surface morphology and evolution on mars: Analog comparisons with tephra-covered glaciers in the valley of ten thousand smokes, alaska. *LPI Contributions*, 3007:3488.
- Dundas, C. M., Bramson, A. M., Ojha, L., Wray, J. J., Mellon, M. T., Byrne, S., McEwen, A. S., Putzig, N. E., Viola, D., Sutton, S., et al. (2018). Exposed subsurface ice sheets in the martian mid-latitudes. *Science*, 359(6372):199–201.
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- Head, J. W., Mustard, J. F., Kreslavsky, M. A., Milliken, R. E., and Marchant, D. R. (2003). Recent ice ages on mars. *Nature*, 426(6968):797–802.
- Laskar, J., Correia, A. C., Gastineau, M., Joutel, F., Levrard, B., and Robutel, P. (2004). Long term evolution and chaotic diffusion of the insolation quantities of mars. *Icarus*, 170(2):343–364.
- Williams, K., Dundas, C., and Kahre, M. (2022). The formation mechanisms for mid-latitude ice scarps on mars. *Icarus*, 386:115174.

## 12 People with knowledge in the subject area

- A minimum of three (3) people must be included. Otherwise, the proposal will not be accepted.
- Neither the Director nor the Co-Director of the project may be included in this list.
- You must inform the people on this list that their names will be included here, as they may eventually be contacted as evaluators of your work. However, please note that they will not necessarily serve as your evaluators.
- All fields must be completed.

|                                                                                   |
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| Full name: Joe Levy                                                               |
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## 13 Signatures

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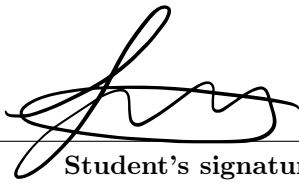
Director's signature

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Co-Director's signature

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Student's signature